

JINYUAN LIU

Department of Mechanical and Aerospace Engineering,
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EDUCATION

Ph. D. Student

Jan. 2021 – present

Department of Mechanical and Aerospace Engineering, UC San Diego
Advisor: Sutanu Sarkar

- Topic: Instabilities, coherent structures, and turbulence in natural and engineering wakes

B. S. E., Engineering Thermo-physics (with distinction)

Aug. 2016 – June 2020

Department of Engineering Mechanics, Tsinghua University

- Dissertation Title: Large- and very-large-scale structures in turbulent channels

Visiting Research Student

Sep. 2019 – Mar. 2020

School of Aeronautics, Universidad Politécnica de Madrid
Advisor: Javier Jiménez

- Topic: Lifetimes of the largest structures in turbulent channels up to $Re_\tau = 5300$.

AWARDS

Academic Distinction Award for Graduates, Tsinghua University	2020
Outstanding Undergraduate Dissertation, Tsinghua University	2020
Second Prize for Student Research Training, Tsinghua University	2019
Scholarship for Scientific Innovation, Tsinghua University	2018
Scholarship for Overall Excellence, Tsinghua University	2018
Scholarship for Academic Distinction, Tsinghua University	2017, 2018, 2019
First Prize, National Aerospace Knowledge Competition for College Students	2018
Outstanding Teaching Assistant of the Year, MAE Department, UC San Diego	2024

FELLOWSHIPS AND MEMBERSHIPS

Fellowship, the fifth Madrid Turbulence Summer School, Universidad Politécnica de Madrid	2023
Fellowship, Spark Innovative Talent Cultivation Program, Tsinghua University	2018
Student Membership, American Physical Society, Division of Fluid Dynamics	since 2019
Student Membership, American Institute of Aeronautics and Astronautics	since 2024

WORK EXPERIENCE

Graduate Research Assistant

Jan. 2021 – present

Mechanical and Aerospace Engineering, UC San Diego
Advisor: Sutanu Sarkar

Post-baccalaureate Research Assistant

July 2020 – Dec. 2020

Department of Engineering Mechanics, Tsinghua University
Advisor: Wei-Xi Huang

Research Assistant

Sep. 2019 – Mar. 2020

School of Aeronautics, Universidad Politécnica de Madrid
Advisor: Javier Jiménez

RESEARCH ACTIVITIES

Publications and preprints

5. Jinyuan Liu and Sutanu Sarkar. Scale interactions and energy transfer in the turbulent wake of a bluff body. *submitted to Journal of Fluid Mechanics*, 2026. [arXiv.2607.10028](https://arxiv.org/abs/2607.10028)
4. Jinyuan Liu, Pranav Puthan, and Sutanu Sarkar. Turbulence in stratified rotating topographic wakes. *Journal of Physical Oceanography*, 55:2159–2177, 2025. doi.org/10.1175/JPO-D-25-0035.1
3. Jinyuan Liu, Pranav Puthan, and Sutanu Sarkar. Effect of rotation on wake vortices in stratified flow. *Journal of Fluid Mechanics*, 999:A44, 2024. [doi:10.1017/jfm.2024.721](https://doi.org/10.1017/jfm.2024.721)
2. Jinyuan Liu and Sutanu Sarkar. Coherence analysis of geophysical wakes in flow past a three-dimensional hill. In *Journal of Physics: Conference Series (Proceeding for the fifth Madrid Turbulence Summer School)*. IOP Publishing, 2024. [doi:10.1088/1742-6596/2753/1/012007](https://doi.org/10.1088/1742-6596/2753/1/012007)
1. Chunmei Xie, Jinyuan Liu, Jing-Wei Jiang, and Wei-Xi Huang. Numerical study on wetted and cavitating tip-vortical flows around an elliptical hydrofoil: Interplay of cavitation, vortices, and turbulence. *Physics of Fluids*, 33(9):093316, 2021. doi.org/10.1063/5.0064717

In prepration

1. Jinyuan Liu and Sutanu Sarkar. On the role of coherent structures in turbulent axisymmetric wakes.
2. Jinyuan Liu and Sutanu Sarkar. Transition to three-dimensionality in strongly stratified wakes.

Talks and conference presentations

15. **Jinyuan Liu** and Sutanu Sarkar. Coherent structures and turbulence in bluff-body wakes. AIAA SciTech Forum 2027, Orlando, January 2027.
14. **Jinyuan Liu** and Sutanu Sarkar. On the role of coherent structures in turbulent wake evolution. The 79th Annual Meeting of APS DFD, Orlando, November 2026.
13. **Jinyuan Liu** and Sutanu Sarkar. Scale-based energy transfer in a turbulent disk wake. The 19th SoCal Fluids Symposium, UC Irvine, Los Angeles, April 2026.
12. **Jinyuan Liu** and Sutanu Sarkar. Inter-scale energy transfer in wake turbulence. Jacob School Research Expo, UCSD, April 2026.
11. **Jinyuan Liu** and Sutanu Sarkar. Interscale energy transfer in turbulent axisymmetric wakes. The 78th Annual Meeting of APS DFD, Houston, November 2025.
10. **Jinyuan Liu**, Pranav Puthan, and Sutanu Sarkar. Turbulence in geophysical wakes: a parametric study. The 18th SoCal Fluids Symposium, USC, Los Angeles, April 2025.
9. **Jinyuan Liu**, Pranav Puthan, and Sutanu Sarkar. Turbulence generated by an isolated topography in stratified rotating flows. The 77th Annual Meeting of APS DFD, Salt Lake City, November 2024.
8. **Jinyuan Liu** and Sutanu Sarkar. Coherent structures and their minimal flow units in strongly stratified Kármán wakes. Workshop on Coherent Structures for Turbulence Modeling, Santa Fe, November 2024.

7. **Jinyuan Liu**, Pranav Puthan, and Sutanu Sarkar. Stratified wakes past an isolated topography. The 17th SoCal Fluids Symposium, UCI, Irvine, April 2024.
6. **Jinyuan Liu**, Pranav Puthan, and Sutanu Sarkar. Stratified wakes past an isolated topography: coherent structures and vortex dynamics. Jacob School Research Expo, UCSD, April 2024.
5. **Jinyuan Liu**, Pranav Puthan, and Sutanu Sarkar. Stratified wakes past an isolated topography: coherent structures and vortex dynamics. Ocean Science Meeting 2024, New Orleans, February 2024.
4. **Jinyuan Liu**, Pranav Puthan, and Sutanu Sarkar. Coherent vortex structures in stratified rotating flow past a three-dimensional hill. The 76th Annual Meeting of APS DFD, Washington DC, November 2023.
3. **Jinyuan Liu**, Pranav Puthan, and Sutanu Sarkar. Influence of rotation on the vortex dynamics in stratified flow past an underwater hill. The 16th SoCal Fluids Symposium, SDSU, San Diego, April 2023.
2. **Jinyuan Liu**, Pranav Puthan, and Sutanu Sarkar. Spatiotemporal analysis of coherent structures in the wake of an underwater hill. The 75th Annual Meeting of APS DFD, Indianapolis, November 2022.
1. **Jinyuan Liu**, Pranav Puthan, and Sutanu Sarkar. Structures of vortices in underwater hill wakes. The 15th SoCal Fluids Symposium, UCLA, Los Angeles, April 2022.

TEACHING ACTIVITIES

Teaching Assistant, UC San Diego	Winters 2022, 2023, 2024
MAE290B Numerical Methods for ODEs & PDEs (graduate level)	
Instructor: Sutanu Sarkar	
Teaching Assistant, UC San Diego	Springs 2022, 2023, 2025, 2026
MAE214A Introduction to Turbulence and Turbulent Mixing (graduate level)	
Instructor: Sutanu Sarkar	
Teaching Assistant, UC San Diego	Winter 2025
MAE101B 'Advanced' Fluid Mechanics (undergraduate level)	
Instructor: Hieu Pham	
Teaching Assistant, UC San Diego	Fall 2026
MAE101A Introductory Fluid Mechanics (undergraduate level)	
Instructor: Sutanu Sarkar	
Teaching Assistant, UC San Diego	Fall 2025
MAE8 MATLAB Programming (undergraduate level)	
Instructor: Hieu Pham	

SERVICES

Referee for: <i>Journal of Fluid Mechanics</i> , <i>Physics of Fluids</i> , <i>Acta Mechanica Sinica</i>	
Chairperson, the 12 th Spark Innovative Talent Cultivation Program, Tsinghua University	
Peer Mentor of 15 freshmen, School of Aerospace, Tsinghua University	2018, 2019
JUMP Mentor of 8 freshmen, Jacob School of Engineering, UC San Diego	2020, 2021

TECHNICAL SKILLS

Programming Languages	FORTRAN, C++, MATLAB, Python, L ^A T _E X, Bash (Unix Shell)
Numerical Packages	MPI, HDF5, FFTW, LAPACK, SCALAPACK, BLAS, FISHPACK
Software Capability	ParaView, Semtex, Nek5000, Dedalus, OpenFOAM
CAE Usage (rarely)	AutoCAD, FreeCAD, SOLIDWORKS, ANSYS ICEM, ANSYS Fluent

RELEVANT COURSEWORK

Fluids	Tsinghua: Fluid Mechanics, Aerodynamics, Compressible Flows, Turbulence Theory, Viscous Fluid Flows, Turbulence Modeling, Heat Transfer, Numerical Heat Transfer, CFD, Advanced CFD, Combustion Theory, Propulsion Theory UCSD: Grad-school Fluid Mechanics, Geophysical Fluid Dynamics (I, III), Ocean Waves, Hydrodynamic Stability, Astrophysical Fluid Dynamics, Turbulence and Mixing
Math	Matrix Analysis, Asymptotic Analysis, Partial Differential Equations, Numerical Analysis, Computational Stochastics, Model Reduction
Numerics	Numerical Linear Algebra, Numerical Methods for ODEs and PDEs, Computational Fluid Dynamics, Parallel Computing

EXTRA-CIRRICULAR

Former member, the baseball team and the water polo team, Tsinghua University	
Former member, the swimming team, School of Aerospace, Tsinghua University	
Go (Wei-Qi) player ranked amateur 5-dan, Chinese Go Association	2013
Honorary Citizen, Tulsa, OK	2014

Updated: Aug, 2026